

End Semester Examination 2025

Name of the Course: B.Tech. Civil Engineering
Name of the Paper: Hydrology and Irrigation Engineering

Semester: V
Paper Code: TCE-504

Time: 3Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each sub question carries 10 marks

Q1	(20marks)								CO 1																	
(a)	Define 'hydrology'. Sketch the block diagram showing "an engineering representation of hydrology", What is the need of studying this subject for Civil Engineers ?																									
(b)	Storm with 10 cm precipitation produced a direct runoff 5.8 cm. Given the time distribution of the storm as below, estimate the ϕ -index of the storm																									
	<table><tr><td>Time from start (hr)</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>Rainfall in each hr. in cm</td><td>0.4</td><td>0.9</td><td>1.5</td><td>2.3</td><td>1.8</td><td>1.6</td><td>1</td><td>0.5</td></tr></table>									Time from start (hr)	1	2	3	4	5	6	7	8	Rainfall in each hr. in cm	0.4	0.9	1.5	2.3	1.8	1.6	1
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Rainfall in each hr. in cm	0.4	0.9	1.5	2.3	1.8	1.6	1	0.5																		
(c)	A reservoir had an average surface area of 20 km ² during June 2025, In that month mean rate of inflow = 10 m ³ /sec mean rate of outflow = 15 m ³ /sec monthly rainfall = 10 cm decrease of storage = 16 Million m ³ Assuming seepage losses to be 18 mm Estimate the evaporation in that month.																									
Q2	(20marks)								CO 2																	
(a)	Define Unit hydrograph and discuss the assumptions and Limitation made in the Unit hydrograph theory.																									
(b)	The peak of flood hydrograph due to 3-hr duration isolated storm in a catchment is 270 cumecs.the total depth of rainfall is 5.9 cm.Assuming an average infiltration loss of 0.3 cm/h and a constant base flow of 20 cumecs and area of catchment is 567 km ² ,estimate i. The peak of the 3-hr unit hydrograph of this catchment The base width of the 3-hr unit hydrograph by assuming it to be triangular in shape																									
(c)	Estimate the maximum flood flow for the following catchment using appropriate empirical formula A = 50 km ² for western ghat area of Maharashtra A = 50 km ² in Gangatic plain A = 50 km ² in Cauvery delta Tamilnadu What is the peak discharge for A = 50 km ² by maximum world flood experience																									
Q3	(20marks)								CO 3																	
(a)	Establish a relation between Duty of water, Delta and Base period. Discuss the factors on which duty of water depends.																									

(b)	Derive the Dupuits formula for a discharge of a well in unconfined aquifer assuming equilibrium flow conditions. State all the assumptions made.																										
(c)	Water requirements of crops during Kharif, Rabi and hot weather seasons in an irrigation scheme are 15, 16 and 10 cumecs respectively. The other data is under: Canal transit losses =15% Capacity factor =0.8 Time factor =0.7 Tank losses=20% Assume base period for above season is 120 days Estimate a) Design discharge of the canal b)The gross storage capacity of the reservoir																										
Q4	(20marks)																										
(a)	Explain with the neat sketch how do you carry out lift irrigation scheme.	CO 4																									
(b)	What are the factors that affect the pattern of sediment deposition in a reservoir?																										
(c)	Following data pertain to monthly flows in river at a particular site. <table><tr><td>month</td><td>Jan</td><td>Feb</td><td>Mar</td><td>April</td><td>May</td><td>June</td><td>July</td><td>Aug</td><td>Sept</td><td>Oct</td><td>Nov</td><td>Dec</td></tr><tr><td>Flow in m³/sec</td><td>70</td><td>55</td><td>45</td><td>30</td><td>25</td><td>35</td><td>60</td><td>90</td><td>115</td><td>100</td><td>90</td><td>80</td></tr></table> Determine the minimum storage required so as to supply constant discharge of 50 m ³ /sec.		month	Jan	Feb	Mar	April	May	June	July	Aug	Sept	Oct	Nov	Dec	Flow in m ³ /sec	70	55	45	30	25	35	60	90	115	100	90
month	Jan	Feb	Mar	April	May	June	July	Aug	Sept	Oct	Nov	Dec															
Flow in m ³ /sec	70	55	45	30	25	35	60	90	115	100	90	80															
Q5	(20marks)																										
(a)	What are the different types of alignment of irrigation canals? Mention the situation where each type is suitable.	CO 5																									
(b)	What are the different types of cross drainage works? State the conditions under which each one is adopted. Sketch any one of them.																										
(c)	Design a channel in allowed soil for the following data by Kennedy methods taking values of the coefficient and exponents in Kennedy formula as 0.55 and 0.64 full supply discharge =55m ³ /sec $\frac{B}{D} = 11$ Side slope= 0.5H:1V CVR (m)=1.0 N=0.0225																										